

Problem 1

- (a) Let A be the set of all functions $f : \{0, 1\} \rightarrow \mathbb{N}$

Let $g : A \rightarrow \mathbb{N} \times \mathbb{N}$ s.t. $g(f) = (f(0), f(1))$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus this means that $(f(0), f(1)) = (h(0), h(1))$

This implies that $f = h$ since they map the same input to the same output for their whole domain thus g is injective.

Surjective:

Let $a, b \in \mathbb{N}$ thus consider $(a, b) \in \mathbb{N} \times \mathbb{N}$

Consider the function f that maps 0 to a and 1 to b . This function is in A . Thus, $g(f) = (a, b)$

Thus g is bijective, thus by proven in class since $\mathbb{N} \times \mathbb{N}$ has a bijection to A we know that A is countable.

- (b) Consider F_n which is all functions from $\{1, \dots, n\}$ to $\mathbb{N}$.

Using a similar proof to (a) we can show that F_n has a bijection to $\mathbb{N}^n$, the Cartesian product of $\mathbb{N}$ taken n -times.

Shown in class that finite number of Cartesian products is countable thus each F_n is countable.

Then proven in class we showed that countable union of countable sets is countable, and since we only take $|\mathbb{N}|$ number of unions this is a countable number thus the set is countable.

- (c) Let A be the set of all functions $f : \mathbb{N} \rightarrow \{0, 1\}$

Let $g : A \rightarrow \{\text{infinite binary sequences}\}$ s.t. $g(f) = (f(1), f(2), \dots)$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus this means that $(f(1), f(2), \dots) = (h(1), h(2), \dots)$ thus $f(1) = h(1), f(2) = h(2), \dots$ thus $f = h$ thus g is injective.

Surjective:

Let $s = (x_1, x_2, \dots)$ be a infinite binary sequence. Thus pick the function f in A s.t. $f(1) = x_1, f(2) = x_2, \dots$ thus $g(f) = s$

Thus g is bijective.

Now assume for contradiction that the set of infinite binary sequences is countable. Thus, we can list them $s_1, s_2, \dots$

Now define the sequence t s.t. $t(n) = 1$ if s_n at the n th term is 0 and 0 if the n th term is 1.

t cannot be in our list because if it were lets say it was s_m but at the m th spot we flipped it in t thus this is a contradiction.

Thus, A is not countable.

- (d) Let A be the set in the question.

Let $g : A \rightarrow \bigcup_{N \in \mathbb{N}} \{\text{binary sequences of length } N - 1\}$ s.t. for f s.t it is eventually zero at $N \in \mathbb{N}$

this means for all $n \geq N$, $f(n) = 0$ thus define g s.t. $g(f) = (f(1), f(2), \dots, f(N-1))$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus we know that both f, h both become zero at the same spot lets call it N and that $f(1) = h(1), \dots, f(N-1) = h(N-1)$ thus these are the same functions since everything after and at N is the same. Thus, $f = h$ thus this function is injective.

Surjective:

Let $s \in \bigcup_{N \in \mathbb{N}} \{\text{binary sequences of length } N-1\}$. WLOG let s be length $N \in \mathbb{N}$. Thus pick the function in A s.t. $f(1) = x_1, f(2) = x_2, \dots, f(N) = x_N$ s.t. x_i are the i th term in s , and that f is zero after and at N

Thus $g(f) = s$

Thus A is bijective to the set of all finite binary sequences. Thus, for each size the number of sequences is 2^N where N is the size, thus each set is finite thus finite then we do a countable union on these finite sets thus this whole set is countable.

- (e) Let A be the set in the question.

Let $g : A \rightarrow \bigcup_{N \in \mathbb{N}} \mathbb{N}^N$ s.t. for f s.t it is eventually constant at $N \in \mathbb{N}$ this means for all $n \geq N$, $f(n) = m$ for $m \in \mathbb{N}$ thus define g s.t. $g(f) = (f(1), f(2), \dots, f(N))$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus we know that both f, h both become constant at the same spot lets call it N and that $f(1) = h(1), \dots, f(N) = h(N)$ thus these are the same functions since everything after and at N is the same. Thus, $f = h$ thus injective. Surjective:

Let $s = (x_1, x_2, \dots, x_N)$. WLOG s is length N . Thus pick the function in A s.t. $f(i) = x_i$ for $i \in \{1, \dots, N\}$, and f is constant at and after N . Thus, $g(f) = s$

Thus g is bijective. Thus, since finite cartesian products are countable we are doing countable unions on countable sets thus the whole set is countable.

- (f) Let A be the set of all two-element subsets of $\mathbb{N}$. Thus $A = \{\{a, b\} \subset \mathbb{N} \mid a \neq b\}$

Let $g : A \rightarrow \{(a, b) \in \mathbb{N} \times \mathbb{N} \mid a < b\}$ s.t. $g(\{a, b\}) = (a, b)$ if $a < b$ or (b, a) if $b < a$.

Injective:

Let $c = \{a, b\}, d = \{e, f\} \in A$ s.t. $g(c) = g(d)$. WLOG let $a < b$ and $e < f$. Thus, since $g(c) = g(d)$ we know that $a = e$ and $b = f$. Thus, we know that c, d are the same subset.

Surjective:

Consider (a, b) consider the two element subset $\{a, b\}$ thus $g(\{a, b\}) = (a, b)$.

Thus, A has a bijection to a subset of a countable set $\mathbb{N} \times \mathbb{N}$ thus it is countable.

- (g) Let A be all finite subsets. Do a similar proof in A , but do the bijection to $\bigcup_{N \in \mathbb{N}} \{(x_1, \dots, x_N) \in \mathbb{N}^N \mid x_1 < \dots < x_N\}$. Thus since we know that finite cartesian products are countable, and we do a countable union on them we know that the whole set is countable.

Problem 2

(a) Reflexivity: Since $f(x) = f(x)$ thus $x \sim x$

Symmetry: If $x \sim y$ then $f(x) = f(y)$ then $f(y) = f(x)$ thus $y \sim x$

Transitivity: If $x \sim y$ and $y \sim z$ then $f(x) = f(y)$ and $f(y) = f(z)$ thus $f(x) = f(z)$ thus $x \sim z$

Thus this is an equivalence relation.

Let $g : \{\text{equivalence classes of } A\} = C \rightarrow B$ s.t. $g([x]) = f(x)$

Injective:

Let $[a], [b] \in C$ s.t. $g([a]) = g([b])$. Thus, we know that $f(a) = f(b)$ thus we know that $a \sim b$ thus $[a] = [b]$ Thus injective.

Surjective:

Let $b \in B$ since f is surjective there exists an $a \in A$ s.t. $f(a) = b$ then take the equivalence class a is in thus $g([a]) = f(a) = b$ thus surjective.

Thus, g is bijective.

(b) Reflexivity: Since $A = A$ we can use the identity map from A to A which is a bijection thus we know that $|A| = |A|$ thus $A \sim A$.

Symmetry: If $A \sim B$ then $|A| = |B|$ thus $|B| = |A|$ thus $B \sim A$

Transitivity: If $A \sim B$ and $B \sim C$ then there is a bijection from A to B and B to C thus compose these bijections and we get a bijection from A to C thus $|A| = |C|$ thus $A \sim C$.

Problem 3

Assume $f : A \rightarrow B$, and $g : B \rightarrow A$ are injections. Let $B' = g(B)$. Since g is an injection we have a left inverse. Just restrict the co-domain of g to B' , and we get a bijection. Call thus $g' : B \rightarrow B'$. Define $f' : A \rightarrow B'$ s.t. $f'(x) = g' \circ f$. This is an injection since g' is a bijection and f is an injection. Now Define $A_1 = A, B_1 = B'$, and for $n > 1$, $A_n = f'(A_{n-1})$ and $B_n = f'(B_{n-1})$. Define $h : A \rightarrow B'$ s.t.

$$h(x) = \begin{cases} f'(x) & \text{if } x \in A_n \setminus B_n \text{ for the some } n, \\ x & \text{otherwise} \end{cases}$$

Base Case:

Show $A_2 \subset B' \subset A$ first we have $B' \subset A$. Since $A_2 = f'(A) \subset B'$ we have that this statement is true.

Inductive Step:

Assume for all $n \leq k \in \mathbb{N}$ we have $A_{n+1} \subset B_n \subset A_n$.

By inductive hypothesis we have that $A_{k+1} \subset B_k$ thus $f(A_{k+1}) \subset f(B_k)$ thus $A_{k+2} \subset B_{k+1}$

By inductive hypothesis we have that $B_k \subset A_k$ thus $f(B_k) \subset f(A_k)$ thus $B_{k+1} \subset A_{k+1}$

Thus by induction $A_{n+1} \subset B_n \subset A_n$, for all $n \in \mathbb{N}$

Well-Defined:

If $x \in A \setminus B'$ then $h(x) = f'(x) \in B'$

Otherwise $x \in B'$, so $h(x) = x \in B'$

Injective:

Let $x, y \in A$ s.t. $h(x) = h(y)$

If $x \in A_k \setminus B_k$ for $k \in \mathbb{N}$, and $y \in A_m \setminus B_m$ for $m \in \mathbb{N}$ then $f'(x) = f'(y)$ and since f' is injective we get that $x = y$

If $x, y \notin A_n \setminus B_n$ for all $n \in \mathbb{N}$ then $x = y$

If $x \in A_k \setminus B_k$ for $k \in \mathbb{N}$, and $y \notin A_n \setminus B_n$ for all $n \in \mathbb{N}$. Thus, we have that $h(y) = y = f'(x)$. Since $x \notin B_k$ by the proof above it is not in B_{k+1} , so $f'(x) \in A_{k+1} \setminus B_{k+1}$. However, $y \notin A_n \setminus B_n$ for any $n \in \mathbb{N}$ thus $f'(x) = y$ is a contradiction. Thus, this is never the case.

Thus injective.

Surjective:

Let $y \in B' \subseteq A$

If $y \in A_k \setminus B_k$ for some $k \in \mathbb{N}$. It cannot be $k = 1$. Thus, $k > 1$. Thus $y \in A_k = f'(A_{k-1})$ thus there exists a $x \in A_{k-1}$ s.t. $f'(x) = y$.

Have to show $x \notin B_{k-1}$. Assume it is. This means that $f'(x) = y \in B_k$ which is a contradiction.

Thus, $x \in A_{k-1} \setminus B_{k-1}$ thus $h(x) = f'(x) = y$

If $y \notin A_n \setminus B_n$ for all $n \in \mathbb{N}$ then $h(y) = y$. Thus, surjective.

Thus, h is bijective. Thus $|A| = |B|$

Problem 4

Let $f : (X^\omega)^\omega \rightarrow X^\omega$ s.t. given $S = (s_1, s_2, \dots)$ s.t. $s_i = (x_{i,1}, x_{i,2}, \dots)$ we can define $f(S) = (y_{\pi(1,1)} = x_{1,1}, y_{\pi(2,1)} = x_{2,1}, \dots)$ s.t. $\pi(i, j) = \frac{(i+j-2)(i+j-1)}{2} + j$ gives the order which is basically diagonal order.

Injective:

Let $a, b \in (X^\omega)^\omega$ s.t. $f(a) = f(b)$ thus we know that all items in the sequence $f(a), f(b)$ are the same. Thus, since the function f will map spot i, j to $\pi(i, j)$ we thus know that each i, j in a, b are the same thus $a = b$ thus they are injective.

Surjective:

Let $s \in X^\omega$ consider the element $a \in (X^\omega)^\omega$ s.t. the (i, j) th spot of a is equal to $x_{\pi(i, j)}$ in s . Thus, $f(a) = s$ thus surjective.

Thus, we have found a bijection thus they have the same cardinality.

Problem 5

- (a) Let X be a well-ordered set. Let A be a bounded above subset of X . Thus consider the set $U = \{u \in X \mid u \geq a \text{ for all } a \in A\}$.

This is non empty since A is bounded above. Thus, since X is well-ordered any nonempty subset has a min thus we know found the least upper bound.

- (b) Let X be a well-ordered set. Let A be a nonempty subset of X . Let B be a nonempty subset of A . Thus, B is also a subset of X . Thus, B has a min. Thus, A is well-ordered.

Consider $\mathbb{R}$ and the subset $\mathbb{Q}$. Consider the subset of $\mathbb{Q}$ $A = \{x \in \mathbb{Q} \mid x < \sqrt{2}\}$. The supremum of this set is $\sqrt{2}$

- (c) Let $(a, b), (c, d) \in A \times B$ with $(a, c) \neq (c, d)$. Assume $(a, b) < (c, d)$. Assume for contradiction that $(c, d) < (a, b)$

Thus, we have that either $a < c$, or $a = c$ and $b < d$, and we have that either $c < a$, or $c = a$ and $d < b$. If $a < c$ then $a \neq c$ thus this implies $c < a$ thus this cannot hold. If $a = c$ and $b < d$ then this also implies $d < b$ which is a contradiction thus if $(a, b) < (c, d)$ then $(c, d) \not< (a, b)$

Let $(a, b) \in A \times B$ then Assume $(a, b) < (a, b)$ then we have that either $a < a$ or $a = a$ and $b < b$ well in both cases there are contradictions from the ordering of A and B thus $(a, b) \not< (a, b)$

Let $(a, b), (c, d), (e, f) \in A \times B$. Assume $(a, b) < (c, d)$ and $(c, d) < (e, f)$.

Thus, either $a < c$ or $a = c$ and $b < d$.

If $a < c$ then if $c < e$ then $a < c < e$ thus $(a, b) < (e, f)$. If $c = e$ then $a < c = e$ thus $a < e$ thus $(a, b) < (e, f)$

If $a = c$ and $b < d$ then if $c < e$ then $a = c < e$ thus $a < e$ thus $(a, b) < (e, f)$. If $c = e$ and $d < f$ then $b < d < f$ thus $b < f$ and $a = e$ thus $(a, b) < (e, f)$.

Thus, that is an order relation.

Assume A, B are well-ordered. Let C be a nonempty subset of $A \times B$

Let $C_A = \{a \in A \mid \exists b \in B \text{ s.t. } (a, b) \in C\}$

Thus since A is well-ordered we can find a min of C_A , and it is nonempty since C is nonempty.

Let α be the mins of C_A . Let $C_B = \{b \in B \mid (\alpha, b) \in C\}$. This is nonempty since C_A is nonempty and that α exists. Thus since B is well-ordered take the min of C_B call it β . Thus, consider (α, β)

For $(c, d) \in C$ s.t. $(c, d) \neq (\alpha, \beta)$ since α is the minimum A in C we know that α is always either less than or equal to c if it is equal then we are done. If they are equal then we know that $\beta \neq d$ since $(\alpha, \beta) \neq (c, d)$. Since β is the min B of all elements of the form (α, f) s.t. $f \in B$ we know that $\beta < d$ thus (α, β) is the min element. Thus, $A \times B$ is well-ordered.

- (d) Let X be a well-ordered set. Let $Y = P(X)$

Let the ordering of Y be for $A, B \in P(X)$ $A < B$ if and only if the least element in $(A \setminus Q) \cup (B \setminus$

A) is in B and not A basically the first element where A and B differ determines the order.

Total order:

Let $A, B \in P(X)$ s.t. $A \neq B$. If they are the same then since we take out what they overlap in the set $(A \setminus B) \cup (B \setminus A)$ the least will only be in one set. Thus either $A < B$ or $B < A$

For A by itself we have that $(A \setminus A) \cup (A \setminus A)$ is empty, so it is undefined at this operation.

Let $C \in P(X)$. Assume $A < B$ and $B < C$. Claim:

$$(A \setminus C) \cup (C \setminus A) = (((A \setminus B) \cup (B \setminus A)) \setminus ((B \setminus C) \cup (C \setminus B))) \cup (((B \setminus C) \cup (C \setminus B)) \setminus ((A \setminus B) \cup (B \setminus A)))$$

Call the left D and the right E .

Let $x \in D$ then it is in A or C but not both.

If x is in A then x is not in C .

If x is in B then since x is not in C thus x is in this set $((B \setminus C) \cup (C \setminus B))$, and since it is in A and B it is not in $((A \setminus B) \cup (B \setminus A))$ thus $x \in (((B \setminus C) \cup (C \setminus B)) \setminus ((A \setminus B) \cup (B \setminus A)))$

If x is not in B then since x is in A and not in C it is in the set $((A \setminus B) \cup (B \setminus A)) \setminus ((B \setminus C) \cup (C \setminus B))$

Now if x is in C this means it is not in A , and we can do a symmetric argument. Thus, $D \subseteq E$

Let $x \in E$. Thus, $x \in ((A \setminus B) \cup (B \setminus A)) = F$, or $((B \setminus C) \cup (C \setminus B)) = G$, but not both.

If $x \in F$ then it is not in G . Thus, we know that x is in A or B , but not both, and from not being in G we know that either it is in $B \cap C$ or neither.

If $x \in A$ then it is not in B then it is not in B and C thus it is in $A \setminus C$ thus it is in D .

If $x \in B$ then it is not in A then it is in $B \cap C$ thus it is in C but not A thus $x \in C \setminus A$ thus in D .

Now if $x \in G$ you can do a symmetric argument thus $E \subseteq D$ thus $E = D$.

Now let m be the least element of $D = E$. We know that m is in A or C , but not both.

Assume for contradiction it is in A this means it is not in C .

This means the least element looking at E is in either F or G , but not both.

If m is in F but not G . This implies it is in $A \setminus B$ which is a contradiction of our assumption that $A < B$.

If m is in G but not F . This means that it is in $B \setminus C$ which is a contradiction of our assumption that $B < C$.

Thus, this implies that $m \in C \setminus A$ thus $A < C$.

Thus, we have a total ordering on the power set.

Assume for contradiction that $f : P(X) \rightarrow X$ is an injection. Consider:

$$D = \{x \in X \mid x \notin f^{-1}(x)\}$$

Since f is injective we can use the fact that it has a left inverse to make this map well-defined, and for $x \notin \text{Img}(f)$ we know that $f^{-1}(x) = \emptyset$ thus it is in D .

Let $f(D) = y$ for some $y \in X$

- If $f(D) = y$, then $f^{-1}(y) = D$, since f is injective.
- If $y \in D$, then $y \notin f^{-1}(y) = D$ which is a contradiction.
- If $y \notin D$, then $y \in f^{-1}(y) = D$ which is a contradiction.

Thus there is an ordered set $P(X)$ which does not inject into X .